

HYDROGENATED NBR: PATENT LANDSCAPE ON SYNTHESIS, HYDROGENATION CONTROL, AND APPLICATION-DRIVEN FORMULATION

1. ABSTRACT

The provided patent set does not center on a single dedicated HNBR synthesis patent family; instead, it spans several application-driven disclosures that use hydrogenated or partially hydrogenated nitrile rubber as a functional elastomer in dispersions, conductive pastes, hoses, and reinforcement treatments. Across the manifest, the target material is consistently defined as NBR-derived rubber in which residual unsaturation is intentionally retained or reduced to a controlled level, typically expressed as residual double bond (RDB) or degree of hydrogenation. The most explicit HNBR-related disclosures are EP3348582B1, US10686185B2, US11131058B2, and US10458573B2, which describe partially hydrogenated nitrile rubber in carbon nanotube dispersions, HNBR-containing cord treatment agents, and HNBR as a hose inner-tube elastomer. EP3178878B1 provides the clearest hydrogenation-process example, but for styrene-butadiene rubber rather than NBR, showing a solution polymerization route followed by catalytic hydrogenation with titanocene dichloride under hydrogen pressure. EP2519350B1 is not an HNBR preparation patent; it is a metathesis-catalyst disclosure that nevertheless mentions depolymerization/hydrogenation of NBR and HNBR as downstream applications. EP2857436B1 and US20220275306A1 are outside the HNBR synthesis domain and mainly provide unrelated polymer/base-oil context. The dominant formulation trend in the HNBR-relevant patents is not new monomer polymerization chemistry, but rather property tuning through hydrogenation level, residual unsaturation, and compounding with solvents, fillers, or conductive media. Reported HNBR use windows emphasize RDB control from 0.5 to 40 wt%, acrylonitrile content around 20 to 50 wt%, and Mooney viscosity ranges of roughly 10 to 120 for dispersion performance. Process-wise, the evidence favors conventional NBR copolymerization followed by hydrogenation, or use of preformed HNBR in downstream formulations, with performance optimized by balancing wettability, viscosity, adhesion, and mechanical reinforcement.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8: - Example Extracted Example Example 9: - Example Extracted Example Example 10: - Example Extracted Example Example 11: - Example Extracted Example Example 12: - Example Extracted Example Example 13: - Example Extracted Example Example 14: - Example Extracted Example Example 15: - Example Extracted Example Example 16: - Example Extracted Example Example 17: - Example Extracted Example Example 18: - Example Extracted Example Example 19: - Example Extracted Example Example 20:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: EP2857436B1
- Patent Title: Polymer having terminal structure including plurality of reactive silicon ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Comparative Example 6: - Example Extracted Example Comparative Example 7: - Example Extracted Example Comparative Example 8: - Example Extracted Example Comparative Example 9: - Example Extracted Example Comparative Example 10: - Example Extracted Example

Comparative Example 11: - Example Extracted Example Comparative Example 12: - Example Extracted Example Comparative Example 13: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: US20220275306A1
- Patent Title: Base oils and methods of making the same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example A: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example B: - Example Extracted Example Example C: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 2: - Example Extracted Example Example 2: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: EP3348582B1
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Comparative Example 6: - Example Extracted Example Comparative Example 7: - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Comparative Example 6: - Example Extracted Example Comparative Example 7: - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Comparative Example 6: - Example Extracted Example Comparative Example 7: - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Comparative Example 6:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US10686185B2
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 54.: - Example Extracted Example Example 47.: - Example Extracted Example Example 53.: - Example Extracted Example Example 43.: - Example Extracted Example Example 65.: - Example Extracted Example Example 43.: - Example Extracted Example Example 54.: - Example Extracted Example Example 43.: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: EP3355392B1
- Patent Title: Carbon nanotube dispersion liquid and manufacturing method thereof
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US20180022951A1
- Patent Title: Electroconductive silver paste
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 2: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 3: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 4: - Example Extracted Example Preparation Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Polymerization / Synthesis Method** - No polymerization parameters disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

- Patent Number: EP3178878B1
- Patent Title: Pneumatic tire
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Synthesis Example A: -
Example Extracted Example Synthesis Example A: - Example Extracted Example Synthesis
Example A: - Example Extracted Example Synthesis Example A: - Example Extracted Example
Synthesis Example A: - Example Extracted Example Synthesis Example B: - Example Extracted
Example Synthesis Example B: - Example Extracted Example Synthesis Example B: - Example
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Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US10458573B2
- Patent Title: High pressure compact spiral hydraulic hose
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP3348582B1 vs US10686185B2: both disclose partially hydrogenated nitrile rubber in CNT dispersions; EP3348582B1 is directed to entangled-type CNTs with D50 about 3-5 μm , while US10686185B2 is directed to entangled-type CNTs with D50 about 2-5 μm . Both specify RDB 0.5-40 wt% and acrylonitrile-derived units at 20-50 wt%, and both describe the rubber as prepared by copolymerizing α,β -unsaturated nitrile with conjugated diene followed by hydrogenation.
- EP3348582B1 vs US11131058B2: EP3348582B1 uses partially hydrogenated nitrile rubber as a CNT dispersant, whereas US11131058B2 uses carboxy-modified hydrogenated acrylonitrile-butadiene copolymer latex in a cord treatment agent. The former emphasizes dispersion/particle-size control; the latter emphasizes adhesion, fraying resistance, and aqueous treatment chemistry.
- US10686185B2 vs US11131058B2: US10686185B2 focuses on dispersion liquid formulation and CNT wetting, with Mooney viscosity 10-120 and RDB 0.5-40 wt%; US11131058B2 focuses on textile cord adhesion and explicitly allows carboxy-modified HNBR latex among the latex options, but does not disclose hydrogenation degree or RDB.

- EP3178878B1 vs the HNBR-focused patents: EP3178878B1 is the only patent in the set with a fully described hydrogenation operation, but it concerns SBR rather than NBR. It uses solution polymerization in n-hexane with n-butyllithium, then hydrogenation at 90°C under 0.7 MPa gauge H₂ using a titanocene dichloride-based catalyst to reach 60-95 mol% hydrogenation. The HNBR patents instead rely on general statements that NBR is copolymerized and then hydrogenated, without comparable operational detail.
- EP2519350B1 vs the HNBR-focused patents: EP2519350B1 is a catalyst/metathesis patent that mentions making modified NBR or depolymerized HNBR via metathesis/depolymerization and subsequent hydrogenation, but it does not disclose an HNBR polymerization recipe. Its role is enabling chemistry rather than elastomer formulation.
- US10458573B2 vs the HNBR-focused patents: US10458573B2 uses HNBR as a hose inner-tube material together with rod-shaped particles, but does not disclose HNBR synthesis. It is application-oriented and contrasts with the dispersion and treatment patents, which specify residual unsaturation, particle size, and latex composition.
- Across the HNBR-relevant patents, the recurring approach is conventional NBR formation followed by controlled hydrogenation, then tailoring of residual unsaturation and acrylonitrile content for end-use compatibility. Newer disclosures emphasize application-specific dispersion stability and adhesion rather than novel hydrogenation catalysts.

4. CONCLUSION

The evidence indicates that hydrogenated NBR is being optimized primarily as a performance elastomer rather than through fundamentally new polymerization chemistry. The most consistent synthesis logic is: prepare NBR by conventional copolymerization of acrylonitrile with conjugated diene, then hydrogenate the diene-derived unsaturation to a controlled residual level. For application performance, the strongest recurring parameter window is acrylonitrile content of about 20-50 wt% and residual double bond content of about 0.5-40 wt%, with Mooney viscosity roughly 10-120 when dispersion processability matters. Where hydrogenation is used in a more explicit process context, the broader patent set supports hydrogenation under hydrogen pressure with a transition-metal catalyst, but the HNBR-specific patents do not provide a single preferred catalyst/pressure/temperature recipe. Based on the available evidence, the most defensible recommended synthesis strategy is conventional NBR copolymerization followed by partial hydrogenation to an RDB in the mid-range of the disclosed window, then formulation-specific adjustment of viscosity, latex form, and surface chemistry to match the target application. For CNT dispersions, lower-to-moderate RDB and acrylonitrile content around 20-50 wt% appear most useful; for cords and hoses, HNBR is used more as a structural or adhesive-compatible elastomer than as a dispersion aid. Overall, the patent landscape favors controlled hydrogenation degree and downstream compounding over novel monomer-feed design.

5. REFERENCES

- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- EP2857436B1 | Polymer having terminal structure including plurality of reactive silicon ... | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2857436B1/en>
- US20220275306A1 | Base oils and methods of making the same | Not disclosed | US | 2022 | <https://patents.google.com/patent/US20220275306A1/en>
- EP3348582B1 | Carbon nanotube dispersion liquid and manufacturing method thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3348582B1/en>
- US10686185B2 | Carbon nanotube dispersion liquid and manufacturing method thereof | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10686185B2/en>
- EP3355392B1 | Carbon nanotube dispersion liquid and manufacturing method thereof | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3355392B1/en>
- US20180022951A1 | Electroconductive silver paste | Not disclosed | US | 2018 | <https://patents.google.com/patent/US20180022951A1/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>
- EP3178878B1 | Pneumatic tire | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3178878B1/en>
- US10458573B2 | High pressure compact spiral hydraulic hose | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10458573B2/en>